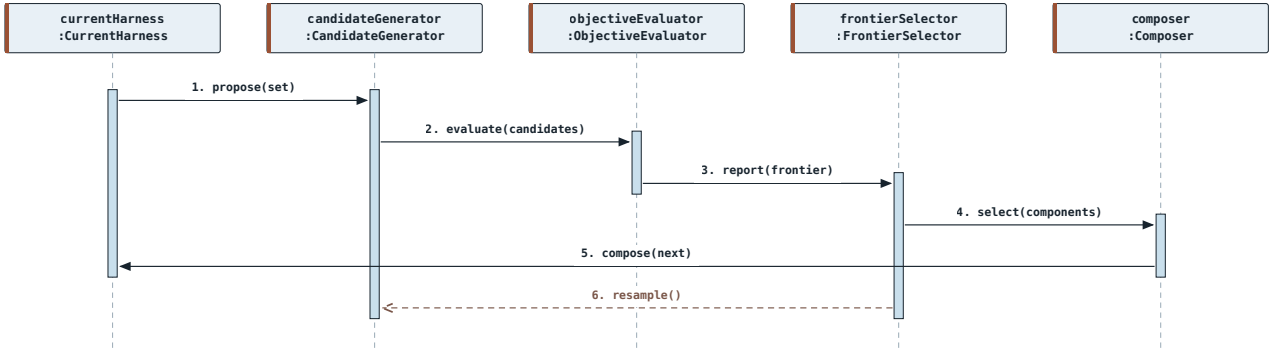
E04 Multi-objective evolution and composition

Evolution / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Search among competing objectives.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Candidate harnesses differ across at least two measured objectives.
TRIGGER	Evaluation yields an objective vector.
STRUCTURE	Declared objectives, constraints, candidate set, and Pareto or scalar selection.
EFFECT	Selected candidates preserve documented trade-offs rather than one hidden score.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Objective direction, weight, and constraints are explicit.
ENSURES	Selection records candidate vectors and dominance or scoring rationale.
INVARIANT	A hard constraint failure cannot be hidden by strong average quality.
VIOLATION	Missing objective or incomparable candidate is marked ineligible or ambiguous.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.